

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think>
- CD21 is a receptor on B cells that is involved in the interaction between B cells and helper T cells .
- EBV and Measles both use CD21 as a receptor to infect B cells .
- HIV and HPV do not use CD21 as a receptor for infection .
- Therefore , the genetic modification that prevents B cells from expressing CD21 would most likely affect the pathogenesis of EBV and Measles , as these viruses rely on CD21 for infection .

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